TOPOLOGICAL INSULATORS — GRADUATE EXAM

Total Points: None | Questions: 5 | Date: April 27, 2026

Diagram-based questions on Topological Insulators

Question 1

10.0 points

Analyze the band structure diagram of a topological insulator shown below, highlighting the non-trivial topology and band inversion. Identify key features such as the gap, conduction and valence bands, and describe the significance of band inversion in the context of topological phases.

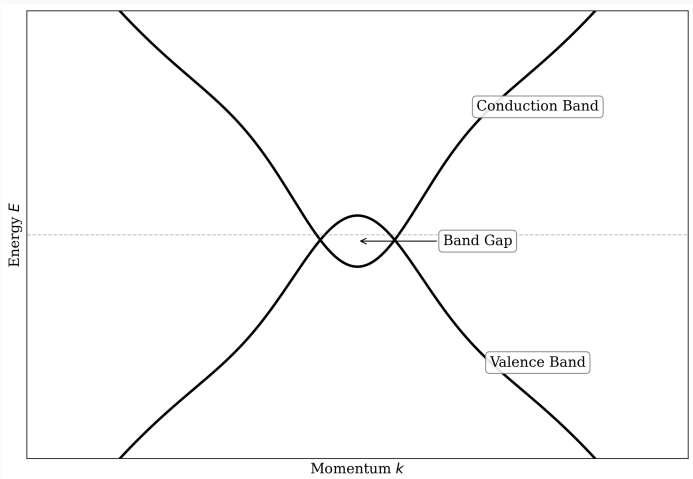

Fig. 1. Band Structure Diagram

Question 2

10.0 points

Explain the concept of bulk-boundary correspondence in topological insulators and its relation to protected surface states, with reference to the diagram below. Discuss the significance of these surface states in a topological insulator.

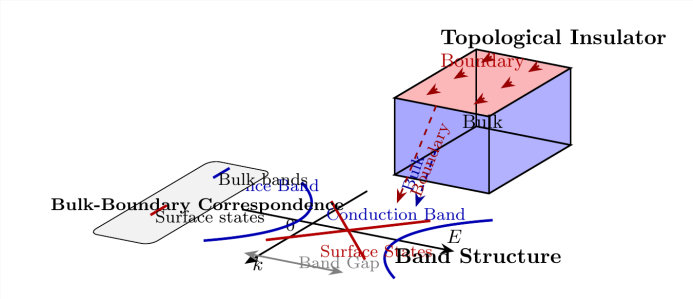

Fig. 2. Topological Insulator Surface States

Question 3

10.0 points

Interpret the quantum spin Hall effect system shown in the diagram below. Highlight the presence of edge states and discuss the role of spin-momentum locking in these states.

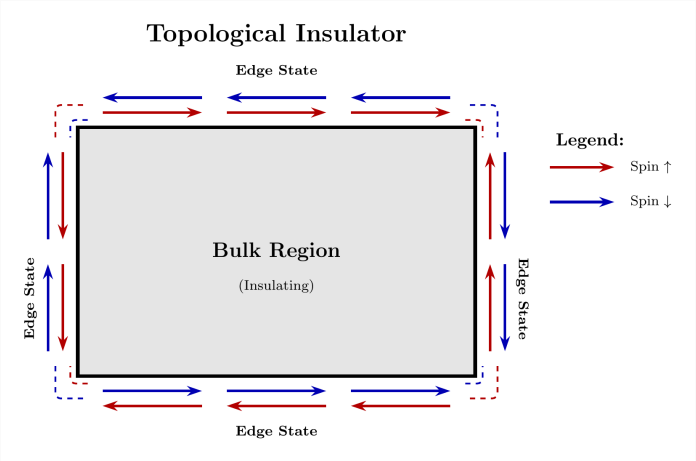

Fig. 3. Quantum Spin Hall Effect

Question 4

10.0 points

Using the k-space diagram below, explain how the Z2 topological invariant is determined for a given material system. Discuss its implications for the topological classification of the material.

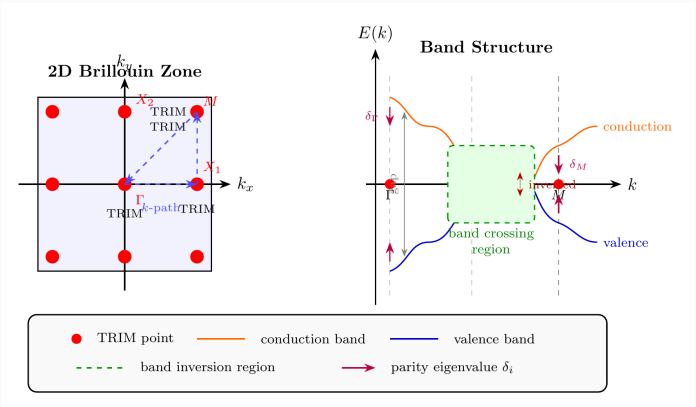

Fig. 4. K Space Diagram

Question 5 10.0 points

Evaluate the diagram below showing the Dirac cone dispersion on the surface of a 3D topological insulator. Describe the significance of the Dirac point and its relationship to surface state protection.

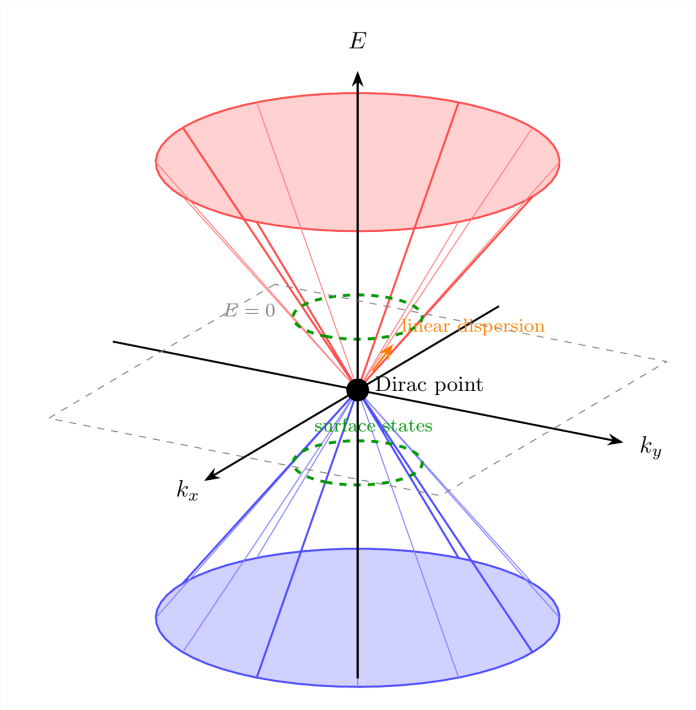

Fig. 5. Dirac Cone Dispersion